

Safety data sheet

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BASF Safety data sheet
Date / Revised: 18.11.2025
Product: **Hydralan® 400 N.V.-1**

Version: 2.0

(30053073/SDS_GEN_VN/EN)

Date of print: 05.12.2025

1. Substance/preparation and manufacturer/supplier identification

Product name:
Hydralan® 400 N.V.-1

Use: brake fluid

Manufacturer/supplier:
BASF Vietnam Co. Ltd.
Level 23, Deutsches Haus, 33 Le Duan,
Sai Gon Ward, Ho Chi Minh City, Vietnam
Telephone: +84 28 3824 3833
Telefax number: +84 28 3824 3832
E-mail address: minh-triet.thieu@basf.com

Emergency information:
18001703 (Vietnam)
Telefax number: +84 28 3824 3832
International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification of the substance and mixture:
Serious eye damage/eye irritation: Cat.2A
Reproductive toxicity: Cat.2 (unborn child)
Reproductive toxicity: Cat.1B (fertility)

Label elements and precautionary statement:

Pictogram:



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Signal Word:
Danger

Hazard Statement:

H319 Causes serious eye irritation.
H360 May damage fertility. Suspected of damaging the unborn child.

Precautionary Statements (Prevention):

P280 Wear protective gloves, protective clothing and eye protection or face protection.
P280 Wear eye protection.
P201 Obtain special instructions before use.
P202 Do not handle until all safety precautions have been read and understood.
P264 Wash contaminated body parts thoroughly after handling.

Precautionary Statements (Response):

P305 + P351 + P338 IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing.
P308 + P313 IF exposed or concerned: Get medical attention.
P337 + P313 If eye irritation persists: Get medical attention.

Precautionary Statements (Storage):

P405 Store locked up.

Precautionary Statements (Disposal):

P501 Dispose of contents and container to hazardous or special waste collection point.

Other hazards which do not result in classification:

If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.
The product does not contain a substance above legal limits fulfilling the PBT (persistent/bioaccumulative/toxic) criteria or the vPvB (very persistent/very bioaccumulative) criteria.

May produce an allergic reaction. Contains:
bisphenol A

3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

Blend based on: polyglycol, glycol ether, inhibitors
, glycoletherborate

Hazardous ingredients

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Ethanol, 2-[2-(2-methoxyethoxy)ethoxy]-, triester with boric acid (H3BO3)	
Content (W/W): $\geq 40\%$ - $\leq 50\%$	Repr.: Cat. 2 (fertility)
CAS Number: 30989-05-0	Repr.: Cat. 2 (unborn child)
2-(2-(2-butoxyethoxy)ethoxy)ethanol	
Content (W/W): $> 10\%$ - $< 20\%$	Acute Tox.: Cat. 5 (dermal)
CAS Number: 143-22-6	Eye Dam.: Cat. 1
2-(2-methoxyethoxy)ethanol	
Content (W/W): $> 1\%$ - $< 3\%$	Flam. Liq.: Cat. 4
CAS Number: 111-77-3	Repr.: Cat. 1B (unborn child)
Diethylene glycol	
Content (W/W): $> 3\%$ - $< 5\%$	Acute Tox.: Cat. 4 (oral)
CAS Number: 111-46-6	
1,1'-iminodipropyl-2-ol	
Content (W/W): $< 3\%$	Eye Irrit.: Cat. 2A
CAS Number: 110-97-4	
bisphenol A	
Content (W/W): $> 0.1\%$ - $< 0.5\%$	Acute Tox.: Cat. 5 (oral)
CAS Number: 80-05-7	Acute Tox.: Cat. 5 (dermal)
	Eye Dam.: Cat. 1
	Skin Sens.: Cat. 1
	Repr.: Cat. 1B (fertility)
	STOT SE: Cat. 3 (irr. to respiratory syst.)
	Aquatic Acute: Cat. 2
	Aquatic Chronic: Cat. 2
3,6,9,12-Tetraoxahexadecan-1-ol	
Content (W/W): $\geq 2\%$ - $\leq 10\%$	Acute Tox.: Cat. 5 (oral)
CAS Number: 1559-34-8	Eye Irrit.: Cat. 2A

4. First-Aid Measures

General advice:

Immediately remove contaminated clothing. If the patient is likely to become unconscious, place and transport in stable sideways position (recovery position). First aid personnel should pay attention to their own safety.

If inhaled:

Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:

Wash thoroughly with soap and water

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On contact with eyes:

Wash affected eyes for at least 15 minutes under running water with eyelids held open, consult an eye specialist.

On ingestion:

Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:

Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far

Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
water spray, dry powder, foam

Specific hazards:
harmful vapours

Generation of fumes/fog. The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear a self-contained breathing apparatus.

Further information:

The degree of risk is governed by the burning substance and the fire conditions. Contaminated extinguishing water must be disposed of in accordance with official regulations.

6. Accidental Release Measures

Personal precautions:

Use personal protective clothing. Breathing protection required.

Environmental precautions:

Contain contaminated water/firefighting water. Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:

For large amounts: Pump off product.

For residues: Pick up with suitable absorbent material. Dispose of absorbed material in accordance with regulations.

7. Handling and Storage

Handling

No special measures necessary provided product is used correctly.

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Protection against fire and explosion:
Take precautionary measures against static discharges.

Storage

Unsuitable materials for containers: Paper/Fibreboard
Further information on storage conditions: Keep container tightly closed and dry; store in a cool place.

8. Exposure controls and personal protection

Components with occupational exposure limits

Ethanol, 2-[2-(2-methoxyethoxy)ethoxy]-, triester with boric acid (H3BO3), 30989-05-0;
TWA value 0.5 mg/m3 (OEL (VN))
STEL value 1 mg/m3 (OEL (VN))

Personal protective equipment

Respiratory protection:

Respiratory protection in case of vapour/aerosol release. Particle filter with medium efficiency for solid and liquid particles (e.g. EN 143 or 149, Type P2 or FFP2)

Hand protection:

Chemical resistant protective gloves

Suitable materials for short-term contact (recommended: At least protective index 2, corresponding > 30 minutes of permeation time according to EN ISO 374-1)

butyl rubber (butyl) - 0.7 mm coating thickness

nitrile rubber (NBR) - 0.4 mm coating thickness

Supplementary note: The specifications are based on tests, literature data and information of glove manufacturers or are derived from similar substances by analogy. Due to many conditions (e.g. temperature) it must be considered, that the practical usage of a chemical-protective glove in practice may be much shorter than the permeation time determined through testing.

Manufacturer's directions for use should be observed because of great diversity of types.

Eye protection:

Tightly fitting safety goggles (splash goggles) (e.g. EN 166)

Body protection:

chemical-protection suit (f.e. according to EN 14605)

General safety and hygiene measures:

Wearing of closed work clothing is required additionally to the stated personal protection equipment.

No eating, drinking, smoking or tobacco use at the place of work. Handle in accordance with good industrial hygiene and safety practice.

9. Physical and Chemical Properties

Form: liquid

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Colour:	yellow	
Odour:	product specific	
Odour threshold:	not determined	
pH value:	7.5 - 9	(FMVSS 116, S 6.4)
solidification temperature:	< -50 °C	(DIN 51583)
boiling temperature:	> 260 °C	(ASTM D1120)
Flash point:	> 130 °C	(DIN 51758)
Evaporation rate:	not determined	
Flammability (solid/gas):	hardly combustible	(derived from flash point)
Lower explosion limit:	For liquids not relevant for classification and labelling., The lower explosion point may be 5 - 15 °C below the flash point.	
Upper explosion limit:	For liquids not relevant for classification and labelling.	
Ignition temperature:	not determined	(DIN 51794)
Thermal decomposition:	No decomposition if correctly stored and handled.	
Self ignition:	not self-igniting	
Explosion hazard:	not explosive	
Fire promoting properties:	not fire-propagating	
Vapour pressure:	1 mbar (20 °C) 1 mbar (50 °C)	
Density:	approx. 1.06 g/cm ³ (20 °C)	
Relative vapour density (air):	> 1 (20 °C) Heavier than air.	(estimated)
Solubility in water:	fully soluble	
Solubility (quantitative) solvent(s):	polar solvents soluble	
Partitioning coefficient n-octanol/water (log Pow):	Study scientifically not justified.	
Viscosity, kinematic:	14 mm ² /s (20 °C)	(DIN 51562)

Other Information:

If necessary, information on other physical and chemical parameters is indicated in this section.

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Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

10. Stability and Reactivity

Conditions to avoid:

Avoid open flames. Avoid humidity.

Thermal
decomposition:

No decomposition if correctly stored and handled.

Substances to avoid:

strong oxidizing agents, strong bases, strong acids

Corrosion to metals: No corrosive effect on metal.

Hazardous reactions:

No hazardous reactions when stored and handled according to instructions.

Hazardous decomposition products:

No hazardous decomposition products known.

Chemical stability:

The product is stable if stored and handled as prescribed/indicated.

11. Toxicological Information

Routes of exposure

Symptoms

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.

(Further) symptoms and / or effects are not known so far

Irritation

Assessment of irritating effects:

Eye contact causes irritation. Not irritating to the skin.

Respiratory/Skin sensitization

Assessment of sensitization:

A sensitizing effect on particularly sensitive individuals cannot be excluded.

Germ cell mutagenicity

Assessment of mutagenicity:

Based on available data, the classification criteria are not met.

Carcinogenicity

Assessment of carcinogenicity:

The whole of the information assessable provides no indication of a carcinogenic effect.

Reproductive toxicity

Assessment of reproduction toxicity:

May impair fertility.

Information on: bisphenol A

Assessment of reproduction toxicity:

The results of animal studies suggest a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:

May cause harm to the unborn child.

Information on: 2-(2-methoxyethoxy)ethanol

Assessment of teratogenicity:

In animal studies the substance caused malformations.

Specific target organ toxicity (single exposure)

Remarks: Based on available data, the classification criteria are not met.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:

Based on available data, the classification criteria are not met.

Aspiration hazard

No aspiration hazard expected.

Other relevant toxicity information

The product has not been tested. The statement has been derived from the properties of the individual components.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:

There is a high probability that the product is not acutely harmful to aquatic organisms.

Mobility

Assessment transport between environmental compartments:

The substance will not evaporate into the atmosphere from the water surface.

Persistence and degradability

Elimination information:

> 70 % DOC reduction (28 d) (OECD 302B; ISO 9888; 88/302/EEC, Teil C) Easily eliminated from water.

Bioaccumulation potential

Assessment bioaccumulation potential:

Significant accumulation in organisms is not to be expected.

13. Disposal Considerations

Information on disposal regulations: Must comply with regulations on hazardous waste such as:

- Decree No. 08/2022/ND-CP of the Government: detailing a number of articles of the Law on Environmental Protection.

- Circular No. 02/2022/TT-BTNMT of the Ministry of Natural Resources and Environment: detailing a number of articles of the Law on Environmental Protection.

Must be disposed of or incinerated in accordance with local regulations.

Contaminated packaging:

Uncontaminated packaging can be re-used.

Packs that cannot be cleaned should be disposed of in the same manner as the contents.

14. Transport Information

Domestic transport:

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
UN proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Sea transport

IMDG

	Not classified as a dangerous good under transport regulations
UN number or ID number:	Not applicable
UN proper shipping name:	Not applicable

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Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
	Marine pollutant: no
Special precautions for user	None known

Air transport

IATA/ICAO

	Not classified as a dangerous good under transport regulations
UN number or ID number	Not applicable
Proper shipping name:	Not applicable
Transport hazard class(es):	Not applicable
Packing group:	Not applicable
Environmental hazards:	Not applicable
Special precautions for user	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

15. Regulatory Information

Other regulations

Legal information on safety, health and environment for chemicals:

Law on Chemicals No. 06/2007/QH12

Chemical Safety Sheets (SDS) are compiled in accordance with international regulations on chemical classification and labelling (GHS) as well as relevant applicable legal regulations:

- QCVN 05A:2020/BCT National Technical Regulation on safety in production, trading, use, preservation and transportation of hazardous chemicals.
- Amendment 1:2024 QCVN 05A:2020/BCT National technical regulation on safety in production, trading, use, preservation and transportation of hazardous chemicals.
- Decree No. 113/2017/ND-CP dated 09/10/2017 of the Government detailing and guiding the implementation of a number of articles of the Law on Chemicals;
- Circular No. 32/2017/TT-BCT dated 28/12/2017 of the Ministry of Industry and Trade specifying and guiding the implementation of a number of articles of the Law on Chemicals and Decree No. 113/2017/ND-CP dated 09/10/2017 of the Government detailing and guiding the implementation of a number of articles of the Law on Chemicals;
- Decree No. 82/2022/ND-CP dated October 18, 2022 of the Government amending and supplementing a number of articles of the Government's Decree No. 113/2017/ND-CP dated October 09, 2017 detailing and guiding the implementation of a number of articles of the Law on Chemicals
- Circular No. 17/2022/TT-BCT dated October 27, 2022 of the Ministry of Industry and Trade amending and supplementing a number of articles of Circular No. 32/2017/TT-BCT dated December 28, 2017 of the Minister of Industry and Trade specifying and guiding the implementation of a

number of articles of the Law on Chemicals and the Government's Decree No. 113/2017/ND-CP dated October 09, 2017 detailing and guiding the exam a number of articles of the Law on Chemicals

- Decree No. 161/2024 ND-CP dated December 18, 2024 of the Government regulating the list of dangerous goods, transportation of dangerous goods and the order and procedures for issuance of licenses and certificates of completion of training programs for drivers or escorts transporting dangerous goods on roads;
- Decree 34/2024/ND-CP dated 31/03/2024 regulating the list of dangerous goods, transporting dangerous goods by road motor vehicles and inland waterway vehicles
- Circular No. 37/2020/TT-BCT dated 30/11/2020 of the Ministry of Industry and Trade stipulating the list of dangerous goods that must be packed in the process of transporting and transporting dangerous goods by road, railway and inland waterway motor vehicles.
- Decree No. 71/2019/ND-CP dated August 30, 2019 of the Government stipulating the sanctioning of administrative violations in the field of chemicals and industrial explosives.

16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.